OPERATING SYSTEMS ORANGE PROBLEM SUBMISSION

Name: Sharanya C Bharadwaj

SRN: PES2UG24CS458

Section: 4H

PHASE 1: A and B

```
sharanya-c-bharadwaj@sharanya-c-bharadwaj-VirtualBox:~/PES2UG24CS458-pes-vcs$ ./test_objects
Stored blob with hash: d58213f5dbe0629b5c2fa28e5c7d4213ea09227ed0221bbe9db5e5c4b9aafc12
Object stored at: .pes/objects/d5/8213f5dbe0629b5c2fa28e5c7d4213ea09227ed0221bbe9db5e5c4b9aafc12
PASS: blob storage
PASS: deduplication
PASS: integrity check

All Phase 1 tests passed.
```

```
sharanya-c-bharadwaj@sharanya-c-bharadwaj-VirtualBox:~/PES2UG24CS458-pes-vcs$ find .pes/objects -type f
.pes/objects/25/ef1fa07ea68a52f800dc80756ee6b7ae34b337afedb9b46a1af8e11ec4f476
.pes/objects/2a/594d39232787fba8eb7287418aec99c8fc2ecdaf5aaf2e650eda471e566fcf
.pes/objects/d5/8213f5dbe0629b5c2fa28e5c7d4213ea09227ed0221bbe9db5e5c4b9aafc12
sharanya-c-bharadwaj@sharanya-c-bharadwaj-VirtualBox:~/PES2UG24CS458-pes-vcs$
```

PHASE 2: A and B

```
sharanya-c-bharadwaj@sharanya-c-bharadwaj-VirtualBox:~/PES2UG24CS458-pes-vcs$ ./test_tree
Serialized tree: 139 bytes
PASS: tree serialize/parse roundtrip
PASS: tree deterministic serialization

All Phase 2 tests passed.
```

```
sharanya-c-bharadwaj@sharanya-c-bharadwaj-VirtualBox:~/PES2UG24CS458-pes-vcs$ find .pes/objects -type f
.pes/objects/25/ef1fa07ea68a52f800dc80756ee6b7ae34b337afedb9b46a1af8e11ec4f476
.pes/objects/2a/594d39232787fba8eb7287418aec99c8fc2ecdaf5aaf2e650eda471e566fcf
.pes/objects/d5/8213f5dbe0629b5c2fa28e5c7d4213ea09227ed0221bbe9db5e5c4b9aafc12
sharanya-c-bharadwaj@sharanya-c-bharadwaj-VirtualBox:~/PES2UG24CS458-pes-vcs$ xxd .pes/objects/25/ef1fa07ea68a52f800dc80756ee6b7ae34b337afedb9b46a1af8e11ec4f476 | head -20
00000000: 626c 6f62 2031 3800 4475 706c 6963 6174  blob 18.Duplicat
00000010: 6520 636f 6e74 656e 740a                 e content.
```

PHASE 3: A and B

```
sharanya-c-bharadwaj@sharanya-c-bharadwaj-VirtualBox:~/PES2UG24CS458-pes-vcs$ make
make: 'pes' is up to date.
sharanya-c-bharadwaj@sharanya-c-bharadwaj-VirtualBox:~/PES2UG24CS458-pes-vcs$ ./pes init
Initialized empty PES repository
sharanya-c-bharadwaj@sharanya-c-bharadwaj-VirtualBox:~/PES2UG24CS458-pes-vcs$ echo "hello" > file1.txt
sharanya-c-bharadwaj@sharanya-c-bharadwaj-VirtualBox:~/PES2UG24CS458-pes-vcs$ echo "world" > file2.txt
sharanya-c-bharadwaj@sharanya-c-bharadwaj-VirtualBox:~/PES2UG24CS458-pes-vcs$ ./pes add file1.txt file2.txt
sharanya-c-bharadwaj@sharanya-c-bharadwaj-VirtualBox:~/PES2UG24CS458-pes-vcs$ ./pes status
Staged changes:
  staged:     file1.txt
  staged:     file2.txt

Unstaged changes:
  (nothing to show)

Untracked files:
  untracked:  object.h
  untracked:  tree.h
  untracked:  pes.h
  untracked:  test_objects
  untracked:  README.md
  untracked:  test_objects.c
  untracked:  test_tree
  untracked:  Makefile
  untracked:  commit.c
  untracked:  object.c
  untracked:  commit.h
  untracked:  test_tree.c
  untracked:  test_sequence.sh
  untracked:  index.c
  untracked:  tree.c
  untracked:  index.h
  untracked:  .gitignore
  untracked:  pes.c
```

```
sharanya-c-bharadwaj@sharanya-c-bharadwaj-VirtualBox:~/PES2UG24CS458-pes-vcs$ cat .pes/index
100664 2cf8d83d9ee29543b34a87727421fdecb7e3f3a183d337639025de576db9ebb4 1776273744 6 file1.txt
100664 e00c50e16a2df38f8d6bf809e181ad0248da6e6719f35f9f7e65d6f606199f7f 1776273761 6 file2.txt
```

PHASE 4: A, B and C

```
sharanya-c-bharadwaj@sharanya-c-bharadwaj-VirtualBox:~/PES2UG24CS458-pes-vcs$ ./pes init
Initialized empty PES repository
sharanya-c-bharadwaj@sharanya-c-bharadwaj-VirtualBox:~/PES2UG24CS458-pes-vcs$ echo "Hello" > hello.txt
sharanya-c-bharadwaj@sharanya-c-bharadwaj-VirtualBox:~/PES2UG24CS458-pes-vcs$ ./pes add hello.txt
sharanya-c-bharadwaj@sharanya-c-bharadwaj-VirtualBox:~/PES2UG24CS458-pes-vcs$ ./pes commit -m "Initial commit"
Committed as f4952ef8aa44077cfda037d4149439140d9ed44418c5fef3fc6559beb84dc04a
sharanya-c-bharadwaj@sharanya-c-bharadwaj-VirtualBox:~/PES2UG24CS458-pes-vcs$ echo "World" >> hello.txt
sharanya-c-bharadwaj@sharanya-c-bharadwaj-VirtualBox:~/PES2UG24CS458-pes-vcs$ ./pes add hello.txt
sharanya-c-bharadwaj@sharanya-c-bharadwaj-VirtualBox:~/PES2UG24CS458-pes-vcs$ ./pes commit -m "Add world"
Committed as e35b17e84e54ec9b6ea39729a80b3d1a3d21443f6ef883de974b92a92df9177e
sharanya-c-bharadwaj@sharanya-c-bharadwaj-VirtualBox:~/PES2UG24CS458-pes-vcs$ echo "Goodbye" > bye.txt
sharanya-c-bharadwaj@sharanya-c-bharadwaj-VirtualBox:~/PES2UG24CS458-pes-vcs$ ./pes add bye.txt
sharanya-c-bharadwaj@sharanya-c-bharadwaj-VirtualBox:~/PES2UG24CS458-pes-vcs$ ./pes commit -m "Add farewell"
Committed as bf6f5c067eda2993d775f2425e5ba6aca736c14f37294243e7470a73282f242a
sharanya-c-bharadwaj@sharanya-c-bharadwaj-VirtualBox:~/PES2UG24CS458-pes-vcs$ ./pes log
commit bf6f5c067eda2993d775f2425e5ba6aca736c14f37294243e7470a73282f242a
Author: PES User <pes@localhost>
Message: Add farewelldd f

commit e35b17e84e54ec9b6ea39729a80b3d1a3d21443f6ef883de974b92a92df9177e
Author: PES User <pes@localhost>
Message: Add worldelldd f

commit f4952ef8aa44077cfda037d4149439140d9ed44418c5fef3fc6559beb84dc04a
Author: PES User <pes@localhost>
Message: Initial commit
```

```
sharanya-c-bharadwaj@sharanya-c-bharadwaj-VirtualBox:~/PES2UG24CS458-pes-vcs$ find .pes -type f | sort
.pes/HEAD
.pes/index
.pes/objects/05/c8d777d0be85748b6788d1b5caf7d69754e6e68060c265ca21cbd6c05e5f43
.pes/objects/25/ef1fa07ea68a52f800dc80756ee6b7ae34b337afedb9b46a1af8e11ec4f476
.pes/objects/2a/594d39232787fba8eb7287418aec99c8fc2ecdaf5aaf2e650eda471e566fcf
.pes/objects/2c/f8d83d9ee29543b34a87727421fdecb7e3f3a183d337639025de576db9ebb4
.pes/objects/66/224663d23e6f4d9de9e2c7e6d8764305a92a3830a1a52d3d5f4aa8007b5c39
.pes/objects/6e/f19b41225c5369f1c104d45d8d85efa9b057b53b14b4b9b939dd74decc5321
.pes/objects/bf/6f5c067eda2993d775f2425e5ba6aca736c14f37294243e7470a73282f242a
.pes/objects/d5/8213f5dbe0629b5c2fa28e5c7d4213ea09227ed0221bbe9db5e5c4b9aafc12
.pes/objects/e0/0c50e16a2df38f8d6bf809e181ad0248da6e6719f35f9f7e65d6f606199f7f
.pes/objects/e3/5b17e84e54ec9b6ea39729a80b3d1a3d21443f6ef883de974b92a92df9177e
.pes/objects/e6/7ed66bcb4a708250a89f315d4d8bb92703343c84df834438880e13a68652bc
.pes/objects/f4/952ef8aa44077cfda037d4149439140d9ed44418c5fef3fc6559beb84dc04a
.pes/refs/heads/main
```

```
sharanya-c-bharadwaj@sharanya-c-bharadwaj-VirtualBox:~/PES2UG24CS458-pes-vcs$ cat .pes/refs/heads/main
bf6f5c067eda2993d775f2425e5ba6aca736c14f37294243e7470a73282f242a
sharanya-c-bharadwaj@sharanya-c-bharadwaj-VirtualBox:~/PES2UG24CS458-pes-vcs$ cat .pes/HEAD
ref: refs/heads/main
```

make test-integration :

```
sharanya-c-bharadwaj@sharanya-c-bharadwaj-VirtualBox:~/PES2UG24CS458-pes-vcs$ make test-integration
=== Running integration tests ===
bash test_sequence.sh
=== PES-VCS Integration Test ===

--- Repository Initialization ---
Initialized empty PES repository
PASS: .pes/objects exists
PASS: .pes/refs/heads exists
PASS: .pes/HEAD exists

--- Staging Files ---
Status after add:
Staged changes:
  staged:     file.txt
  staged:     hello.txt

Unstaged changes:
  (nothing to show)

Untracked files:
  (nothing to show)


--- First Commit ---
Committed as e18ecd46793b5dcd79d393f897b6577fbd36a128c5c80187365a37041685fb1b

Log after first commit:
commit e18ecd46793b5dcd79d393f897b6577fbd36a128c5c80187365a37041685fb1b
Author: PES User <pes@localhost>
Message: Initial committial commiQ


--- Second Commit ---
Committed as 74981530daa971bba8fc8a882b547abab6a0c28410fffac72ff9cb7ffbf610ee

--- Third Commit ---
Committed as 0cdae0c2910c77842428d1ac4e3319553ada8ec4effdc9231e3b45b50f51f394
```

```
--- Full History ---
commit 0cdae0c2910c77842428d1ac4e3319553ada8ec4effdc9231e3b45b50f51f394
Author: PES User <pes@localhost>
Message: Add farewelldd f

commit 74981530daa971bba8fc8a882b547abab6a0c28410fffac72ff9cb7ffbf610ee
Author: PES User <pes@localhost>
Message: Update file.txtf

commit e18ecd46793b5dcd79d393f897b6577fbd36a128c5c80187365a37041685fb1b
Author: PES User <pes@localhost>
Message: Initial commit


--- Reference Chain ---
HEAD:
ref: refs/heads/main
refs/heads/main:
0cdae0c2910c77842428d1ac4e3319553ada8ec4effdc9231e3b45b50f51f394

--- Object Store ---
Objects created:
8
.pes/objects/0b/d69098bd9b9cc5934a610ab65da429b525361147faa7b5b922919e9a23143d
.pes/objects/0c/dae0c2910c77842428d1ac4e3319553ada8ec4effdc9231e3b45b50f51f394
.pes/objects/58/a67ed1c161a4e89a110968310fe31e39920ef68d4c7c7e0d6695797533f50d
.pes/objects/6e/f19b41225c5369f1c104d45d8d85efa9b057b53b14b4b9b939dd74decc5321
.pes/objects/74/981530daa971bba8fc8a882b547abab6a0c28410fffac72ff9cb7ffbf610ee
.pes/objects/b1/7b838c5951aa88c09635c5895ef7e08f7fa1974d901ce282f30e08de0ccd92
.pes/objects/db/07a1451ca9544dbf66d769b505377d765efae7adc6b97b75cc9d2b3b3da6ff
.pes/objects/e1/8ecd46793b5dcd79d393f897b6577fbd36a128c5c80187365a37041685fb1b

=== All integration tests completed ===
```

Some branch related screenshots:

```
sharanya-c-bharadwaj@sharanya-c-bharadwaj-VirtualBox:~/PES2UG24CS458-pes-vcs$ ./pes branch test
Created branch 'test'
sharanya-c-bharadwaj@sharanya-c-bharadwaj-VirtualBox:~/PES2UG24CS458-pes-vcs$ ls .pes/refs/heads
main  test
sharanya-c-bharadwaj@sharanya-c-bharadwaj-VirtualBox:~/PES2UG24CS458-pes-vcs$ cat .pes/refs/heads/test
fc6237a3023d0d0f9b744789ab9f527e2028956829e4c6186fed30ceefdd587e
```

```
sharanya-c-bharadwaj@sharanya-c-bharadwaj-VirtualBox:~/PES2UG24CS458-pes-vcs$ ./pes branch
main
test
```

```
sharanya-c-bharadwaj@sharanya-c-bharadwaj-VirtualBox:~/PES2UG24CS458-pes-vcs$ ./pes checkout test
Switched to 'test'
sharanya-c-bharadwaj@sharanya-c-bharadwaj-VirtualBox:~/PES2UG24CS458-pes-vcs$ cat .pes/HEAD
ref: refs/heads/test
sharanya-c-bharadwaj@sharanya-c-bharadwaj-VirtualBox:~/PES2UG24CS458-pes-vcs$ make
gcc -Wall -Wextra -O2 -c pes.c -o pes.o
gcc -o pes object.o tree.o index.o commit.o pes.o -lcrypto
sharanya-c-bharadwaj@sharanya-c-bharadwaj-VirtualBox:~/PES2UG24CS458-pes-vcs$ ./pes branch -d test
Deleted branch 'test'
sharanya-c-bharadwaj@sharanya-c-bharadwaj-VirtualBox:~/PES2UG24CS458-pes-vcs$ ./pes branch
main
```

PHASE 5 :

Q5.1: Branching and Checkout

In PES-VCS, a branch is stored as a file in .pes/refs/heads/ containing a commit hash.

To implement `pes checkout <branch>`:

- Read commit hash from .pes/refs/heads/<branch>
- Update .pes/HEAD to: ref: refs/heads/<branch>
- Load the commit, get its tree, and restore files in the working directory

This operation is complex because:

- Files may need to be created, modified, or deleted
- Directory structures must be rebuilt from tree objects
- Uncommitted changes may be overwritten

In our implementation, checkout only updates HEAD and does not modify the working directory.

Q5.2: Dirty Working Directory Detection

To detect conflicts during checkout:

- Compare working directory vs index:
- Compute current file hash and compare with index hash --> if different, file is modified
- Compare index vs target commit tree:
- If same file exists but hashes differ --> conflict

Checkout must be refused if file is modified in working directory AND file differs in the target branch.

Q5.3: Detached HEAD

Detached HEAD means .pes/HEAD contains a commit hash directly instead of a branch reference.

- Commits can still be created
- But they are not attached to any branch

Risk:

- These commits may become unreachable and get deleted

Recovery:

- Create a branch pointing to that commit using: pes branch <name>

PHASE 6:

Q6.1: Garbage Collection

Algorithm to remove unreachable objects:

  - Start from all references (HEAD and branches)
  - Traverse commits and follow parent links
  - Traverse tree objects and collect all blobs and subtrees
  - Store reachable hashes in a hash set
  - Delete objects not present in the set

Data structure:

- Hash set for efficient lookup

Estimate:

- For 100,000 commits, around 300,000 to 1,000,000 objects may be visited

Q6.2: GC Race Condition

If GC runs during a commit:

- Commit creates new objects but has not yet updated HEAD
- GC does not see these objects as reachable
- GC deletes them
- Commit finishes and references deleted objects → repository corruption

Git avoids this by:

- Using atomic writes (temporary files + rename)
- Locking references
- Running GC only when safe

GitHub Commits:

```
sharanya-c-bharadwaj@sharanya-c-bharadwaj-VirtualBox:~/PES2UG24CS458-pes-vcs$ git log --oneline
8ab88db (HEAD -> main, origin/main) Phase 5: Final testing
5055889 Phase 5: HEAD switching
2312e42 Phase 5: Checkout implementation
1277649 Phase 5: Branch listing
6c10b4c Phase 5: Branch creation
8bcbebb Phase 4: Commit traversal
76ed172 Phase 4: Log command
854c6c6 Phase 4: HEAD update
6bb67e9 Phase 4: Parent linking
4029311 Phase 4: Commit creation
a7226db Phase 3: Staging improvements
ec93ec4 Phase 3: Metadata handling
7729f87 Phase 3: Index status
8efbd2b Phase 3: Index add
9462a46 Phase 3: Index load/save
38367dd Phase 2: Tree validation
6940282 Phase 2: Build tree from index
ae01c96 Phase 2: Deterministic ordering
3561637 Phase 2: Tree parsing
da80bb5 Phase 2: Tree serialization
52b22f5 Phase 1: Add integrity check
92d1dbc Phase 1: Add deduplication
101aac1 Phase 1: Implement object read
f76eede Phase 1: Add hashing
3197320 Phase 1: Implement object write
2db2eef Add gitignore
f1b45c4 Completed PES-VCS implementation
```